

READING PASSAGE 3

Deep Sea Ecosystems and Adaptation

A The study of deep sea ecosystems and adaptation has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The intellectual lineage of this field extends considerably further than many realise. While modern research techniques have only recently been applied to these questions, philosophers and natural scientists have grappled with related issues for centuries. The transition from speculative philosophical inquiry to systematic empirical investigation represents one of the most significant developments in the intellectual history of this domain.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F Looking ahead, the field stands at a critical juncture. Emerging technologies offer unprecedented opportunities for data collection and analysis, potentially enabling researchers to address questions that have long remained beyond the reach of empirical investigation. At the same time, the accelerating pace of social and technological change means that the phenomena under study are themselves evolving, requiring theoretical frameworks that can accommodate dynamism and discontinuity rather than assuming stability and equilibrium.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 1 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 2

Choose the correct letter, A, B, C or D.

- 2 The author suggests that early studies in this field were limited by:
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes

Question 3

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 3 Dynamic _____ processes can amplify or dampen initial effects in complex systems.

Questions 4–6

Choose the correct letter, A, B, C or D.

- 4 What finding emerged from the landmark study discussed in the passage?
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone
- 5 According to the passage, what is one of the key factors identified by researchers?
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations
- 6 What finding emerged from the landmark study discussed in the passage?
- A. excessive funding
 - B. theoretical and methodological limitations
 - C. public interest
 - D. international collaboration

Question 7

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 7 The passage identifies power _____ as a key factor at the organisational level.

Question 8

Choose the correct letter, A, B, C or D.

- 8 The author suggests that early studies in this field were limited by:
- A. excessive research funding
 - B. translating academic findings into practical policy
 - C. lack of public interest
 - D. too many international collaborations

Question 9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Questions 10–11

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 10 The author believes that all research questions in this field have been answered.
- 11 Early researchers had access to the same technologies as modern scholars.

Question 12

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 12 The landmark study analysed data spanning a period of _____ years.

Question 13

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 13 All critics agree that the research findings are universally applicable.